

Alif Ayman

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WORK EXPERIENCE

Mechanical Engineering Student

May 2025 – Aug. 2025

Kraken Robotic Systems

Mt. Pearl, Canada

- Designed and assembled a SAS electronics test bench to streamline hardware debugging and validation
- Developed SolidWorks components applying DFM/DFA principles to improve manufacturability
- Managed CAD revisions in SolidWorks PDM to ensure configuration control across concurrent projects
- Ran SolidWorks thermal and Ansys FEA simulations to validate designs against performance specs pre-build
- Produced GD&T manufacturing drawings to support accurate prototyping and first-pass fabrication

Innovation and Prototyping Co-op

Sep. 2025 – Dec. 2025

Student Design Hub, Memorial University

St. John's, Canada

- Led design reviews for 15+ teams, flagging manufacturability, tolerance, and structural risks before fabrication
- Produced GD&T drawings and CAM-ready models for CNC, composites, and additive manufacturing
- Conducted Root Cause Analysis (RCA) on prototypes to resolve defects and prevent recurrence in production
- Coordinated fabrication schedules for 600+ members to meet competition and sponsor deadlines
- Organized student design showcases and sponsor outreach events that helped secure \$350K+ in funding

STUDENT TEAMS

Valiant Aerotech – AEAC UAS Competition 2026 (3rd Place)

Jun. 2025 – Present

Mechanical Team Lead

- Led mechanical design of a quadcopter fire-surveillance UAV, coordinating airframe, propulsion, and payload integration
- Designed a composite airframe in SolidWorks using DFM/DFA to minimize weight while enabling rapid assembly
- Sized propulsion system to achieve a 4:1 thrust-to-weight ratio, balancing payload capacity and flight endurance
- Tuned mechanical PID parameters and conducted vibration analysis in MATLAB Simulink to improve flight stability
- Manufactured carbon-fiber and metal AM components in-house and resolved stiffness and deflection issues
- Coordinated mechanical design decisions with avionics and controls teams to ensure system-level integration

Sidus Robotics – Canadian International Rover Challenge

Feb. 2025 – Present

Robotic Arm & Suspension Team

- Reduced robotic arm mass by 20% using Abaqus FEA while preserving joint stiffness and positioning accuracy
- Performed tolerance-driven CAD and kinematic analysis to ensure reliable reach and payload motion
- Designed and tested rocker-bogie suspension components enabling traversal of 15 cm obstacles
- CNC-machined and 3D-printed mechanical linkages and housings for subsystem validation
- Collaborated with electrical and software teams to ensure mechanical interfaces met integration requirements

TECHNICAL SKILLS

CAD & Design: SolidWorks, Inventor, Fusion 360, SolidWorks PDM, DFM/DFA, GD&T

Analysis & Simulation: Abaqus FEA, Ansys Fluent (CFD), MATLAB Simulink

Fabrication & Manufacturing: CNC Machining, FDM/SLA 3D Printing, Composite Fabrication

Software & Tools: Python, MATLAB, NetSuite ERP, Microsoft Office Suite, CoLab

Certifications: SolidWorks CSWA, WHMIS, RPAS Advanced Drone Pilot Licence (Transport Canada)

EDUCATION

Memorial University of Newfoundland

St. John's, Canada

Term 6, Mechanical Engineering, Class of 2028

Jan. 2023 – Present

Relevant Courses: Machine Design, Mechanics of Solids, Fluid Mechanics, Mechatronics, Materials Engineering, Thermodynamics, Production Technology, Engineering Economics, Engineering Graphics & Design

VOLUNTEERING

FTC Robot Inspector & Referee

Dec. 2025

First Robotics NL

St. John's, Canada

- Enforced strict safety code compliance by auditing mechanical systems prior to operational approval
- Evaluated robot designs against competition engineering rules prior to match approval
- Worked with teams to diagnose build issues and resolve technical disputes during matches